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Automatically Identifying Electrode Reaction Mechanisms Using Deep Neural Networks

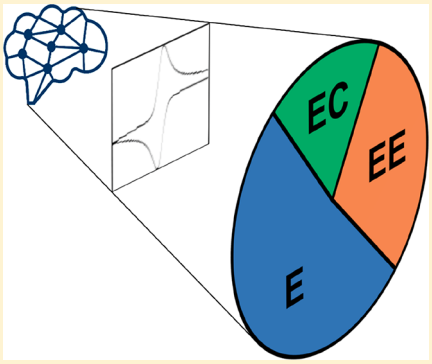
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Supporting Information

ABSTRACT: At present, electrochemical mechanisms are most commonly identified subjectively based on the experience of the researcher. This subjectivity is reflected in bias to particular mechanisms as well as lack of quantifiable confidence in the chosen mechanism compared to potential alternative mechanisms. In this paper we demonstrate that a deep neural network trained to recognize dc cyclic voltammograms for three commonly encountered mechanisms provides correct classifications within 5 ms without the problem of subjectivity. To mimic experimental data, the impact of noise, uncompensated resistance, and dependence on scan rate, factors that are relevant to practical studies, has also been investigated. Outcomes with two experimental data sets are also presented.



Dynamic electrochemistry (voltammetry) has routinely been used to understand the complex phenomena that govern the operation of a wide range of devices relevant for example to electrochemical energy conversion, biomedicine, batteries, fuel cells, and glucose biosensors that underpin important areas of modern society. Voltammetry requires the interpretation of potential−current−time relationships to obtain quantitative mechanistic information about the electrode processes. This has often been undertaken with a heuristic approach involving a series of steps: (1) inspection of voltammetric patterns, (2) identification of the "most likely" reaction mechanism, (3) simulation of theoretical data using different combination of parameters related to kinetics, thermodynamics, and mass transport (together with other general parameters, such as uncompensated resistance, double layer capacitance, electrode area, etc.), and (4) undertaking theory experiment comparison to determine the parameters based on the "best fit". This process is proceeded iteratively until an "acceptable agreement" between theory and experiment is reached. This may also involve the choice of an alternative reaction mechanism if an acceptable agreement cannot be found with the mechanism previously chosen. There are several major drawbacks associated with this heuristic-based approach: (1) tedious and time-consuming, (2) often has a lack of a statistical basis when determining the "best fit" or when "acceptable agreement" has been achieved leading to a significant doubt over the uniqueness of the solution, and (3) subjective: the outcome is dependent on the experimenter. Explicitly, interpretation of voltammetry requires the experimenter to have substantial knowledge of the technique in order to interpret the data correctly. There is no shortage of major mistakes in the history of dynamic electrochemistry due to the incorrect interpretation of the data by highly experienced electrochemists. For example, in the early literature, the reversible ferrocene oxidation process in acetonitrile media detected by transient cyclic voltammetry at a scan rate of 0.1 V s^{-1} was often reported to be quasi-reversible. However, it is reversible when the effect of the uncompensated resistance is taken into consideration.[1]

In view of the drawbacks with heuristic forms of data interpretation, automatically analyzing voltammetric data to obtain quantitative mechanistic information has to be a major objective in modern electrochemistry. The brute force approach of running a very large number of simulations with many free parameters and comparing the simulated result to the experimental data is attractive with high speed computers but remains computationally very intensive so any approach that can narrow the choice of mechanism and the parameter search range is highly desirable.[2−4] Ideally, raw experimental data is automatically analyzed and classified into likely mechanisms with a measure of the confidence as well as any alternative mechanisms that with follow-up experiments could be confirmed or dismissed. Early studies by Perone and co-workers have demonstrated the power of computers to recognize voltammetric patterns.[5−7]

In this paper, we introduce an automatic cyclic voltammetric classifier that quantifies the confidence in each mechanism. The approach taken here is to mimic what an experienced electrochemist does by examining the key features of cyclic voltammograms and classifying the mechanism based on past

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training. Such a task is well suited to deep neural networks (DNNs)[8] in an image recognition context.

DNNs and more generally artificial neural networks have been used in many branches of science with great success, e.g., in biology and medicine,[9] material sciences,[10] multimessenger astrophysics,[11] and crystography.[12] In the field of electrochemistry, artificial neural networks have been used in battery and electrocatalysis research, corrosion studies, and electroanalysis.[13,14] However, an extensive survey of literature reveals a striking lack of activity in the field of dynamic electrochemistry (voltammetry), mainly attributed to the complexity of the voltammetric data. DePalma and Perone determined heterogeneous kinetic parameters from voltammetric data by computerized pattern recognition.[5] Another such study was reported by Sapozhnikova et al. in 2006[15] but was very limited by the state-of-the-art at the time. Basically, these authors used image processing techniques (e.g., wavelets, down-sampling using principal component analysis) to make the problem more tractable and fed the key features into a very simple neural network. They used a range of classifications and hand-picked parameter values to simplify the data analysis. The concept was extended later for estimation of diffusion coefficients from voltammetric data.[16]

The work presented in this paper reveals what can be achieved by applying machine learning in voltammetry using DNN. Three of the most common electrode reaction mechanisms are investigated by dc voltammetry to demonstrate that DNNs can be used to quantify the classification of plausible mechanisms automatically and remove the subjectivity. Our aim is to show that such an approach is perfectly reproducible, quantifiable, and easy to integrate into a data flow beginning with the raw experimental data and ending in a ranked order of mechanisms with associated optimally fitted parameters. To mimic experimental data, the impact of noise, uncompensated resistance, and dependence on scan rate, factors that are relevant to practical studies, has been investigated. Outcomes from the analysis of two experimental data sets are also reported.

RESULTS

To demonstrate the application of DNNs to automatically classifying electrochemical mechanisms we have chosen three widely encountered mechanisms. The first mechanism is a simple one-electron transfer reaction of the form

$$A + e^- \rightleftharpoons B$$

quantitatively defined by a reversible potential (E^0), the heterogeneous electron transfer rate constant at E^0 (k^0), and the charge transfer coefficient (α) according to the Butler–Volmer model of electron transfer. This mechanism is referred to as an E mechanism and is the basis for the other two. In the work of Sapozhnikova et al.[15] the E step was subclassified into reversible and quasi-reversible categories.

The second mechanism (EE) consists of an additional electron transfer reaction described in the reaction scheme of the form

$$A + e^- \rightleftharpoons B; \quad B + e^- \rightleftharpoons D$$

An additional three parameters are needed to define the second E reaction.

The final mechanism under consideration (EC) is a single electron transfer followed by an irreversible chemical reaction (C) that removes the product and acts to suppress the backward electron transfer and has the form

$$A + e^- \rightleftharpoons B; \quad B \rightarrow F$$

The C step is quantified by the forward (k_f) homogeneous chemical reaction rate constant, assuming that this step is fully irreversible.

Table 1. Electrochemical Parameters and Ranges for Generating the DNN Inputs Using MECSim Software

	value/range	default value
Constant Parameter		
temperature	300 K	
$E_{initial}$, $E_{reverse}$, E_{final}	0.5 V, −0.5 V, 0.5 V	
planar electrode area	1 cm^2	
concentration of chemical species [A]	1.0×10^{-5} mol/cm^3	
diffusion coefficient of A, D_A	1.0×10^{-5} cm^2/s	
Variable Parameter		
uncompensated resistance, R_u	10^{-4} to 500 Ω[a]	0.1 Ω
scan rate, ν	10^{-4} to 1 V/s[a]	0.1 V/s
diffusion coefficients for all other species, D_i	10^{-6} to 2×10^{-5} cm^2/s	10^{-5} cm^2/s
double layer capacitance	0–10^{-4} F/cm^2[b]	10^{-5} F/cm^2
Variable Parameter for A + e^- = B		
reversible potential, E^0	−0.3 V to 0.3 V	0.0 V
heterogeneous electron transfer rate constant at E^0, k^0	10^{-4} to 0.5 cm s^{-1}[a]	10^{-3} cm s^{-1}
charge transfer coefficient, α	0.3–0.7	0.5
Variable Parameter for B + e^- = D		
reversible potential, E^0	−0.3 V to 0.3 V	0.0 V
heterogeneous electron transfer rate constant at E^0, k^0	10^{-4} to 0.5 cm s^{-1}[a]	0.0 cm s^{-1}
charge transfer coefficient, α	0.3–0.7	0.5
Variable Parameter for B → F		
forward reaction rate	10^{-2} to 10^3 s^{-1}[a]	0.0 s^{-1}
backward reaction rate	10^{-10} s^{-1}	

[a]Denotes those that are chosen randomly logarithmically rather than linearly. [b]In real experiments, double layer capacitance may not be a constant (independent of potentials).

As the aim here is to demonstrate the applicability of DNNs for classifying these mechanisms based on image recognition, the parameters associated with these images are varied over a large range to provide an adequate size of data for training to cover all scenarios encountered experimentally. Using software such as the Monash Electrochemistry Simulation Package (MECSim) can generate a mathematical model that depends on specified input parameters and allow us to sample a large parameter space. This increases the accuracy of the DNN, as such classifiers work best when they are trained on a large and diverse data set. In practice, experimental issues, such as noise, can bias the DNN. The effect of noise on the performance of the DNN on classifying experimental data in the real world is discussed below.

In addition to the kinetic and thermodynamic parameters mentioned above, this simulation approach is also used to mimic the other parameters that influence the cyclic voltammetric characteristics, such as the diffusion coefficients, uncompensated resistance (R_u, which can have significant influence on the voltammetric patterns if iR_u is substantial), and double layer capacitance (treated as a constant).[17] The full list of parameter ranges is given in Table 1. Note that the conclusions presented here are not parameter dependent, as it

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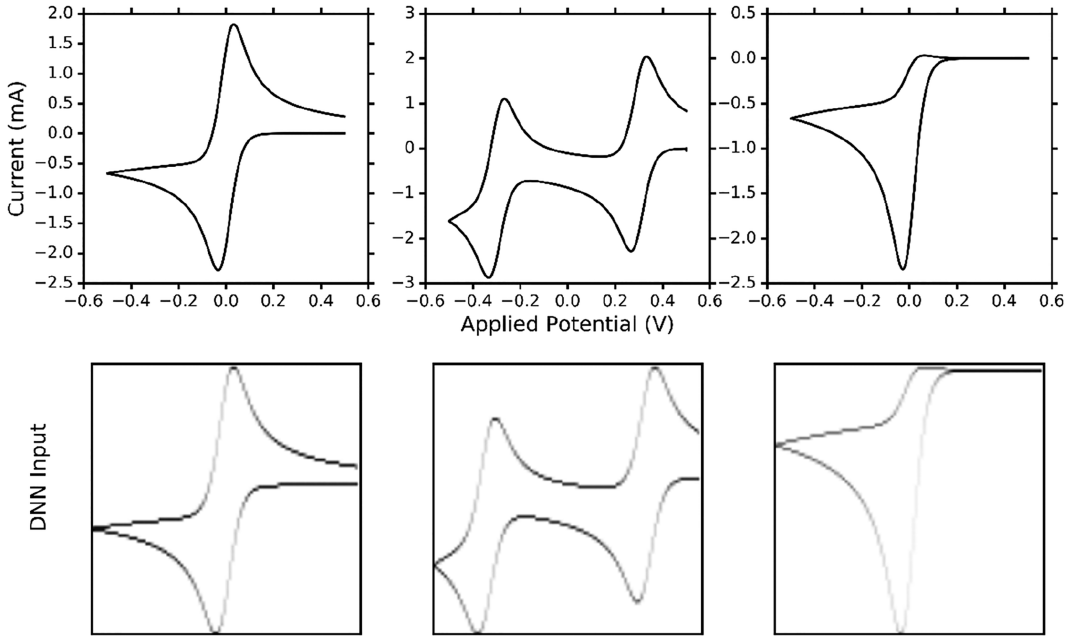


Figure 1. Process for converting the cyclic voltammograms (1st row) into input images for the DNN (2nd row). Columns show the processes for an E, EE, and EC mechanism, respectively, with default parameters given in Table 1 except for $k^0 = 0.1$ cm s^{-1}. Parameters used specifically for the EE and EC mechanisms are $E^0_1 = 0.3$ V, $E^0_2 = -0.3$ V, and $k_f = 10$ s^{-1}, respectively. Note that image resolution in the peak regions decreases more significantly than other regions after conversion into images.

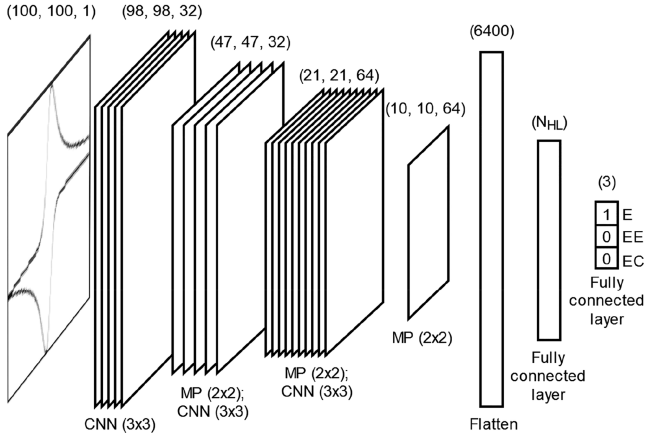


Figure 2. Schematic diagram for the DNN architecture. The initial training data is a normalized greyscale image of 100 × 100 pixels which goes through a series of convolution neural networks (CNN) consisting of 32, 32, and 64 filters where each filter is a 3 × 3 matrix. After each convolution layer, a maximum pooling layer (MP) is developed which takes the maximum value from a 2 × 2 grid and therefore halves the image size. The image sizes at the start of each CNN or MP layer are shown above the layers in the image. The final pooling layer results in a stack of 64 10 × 10 pixel grids which is then flattened to a single 1D array of 6400 nodes. Each node from this layer is fully connected to all the nodes in the hidden layer (e.g., 70 nodes) which are then fully connected to the 3 nodes in the output layer. During training, the output layer values are known and the CNN parameters and weights for connecting the nodes are fitted. During prediction, the fitted parameters and weights are used to determine the values in the output layer corresponding to the probabilities that the input mechanism is an E, EE, or EC mechanism.

is the shape of the cyclic voltammogram that ultimately matters. The parameters and ranges are chosen to reflect the diversity of shapes for each mechanism.

Table 2. List of Hyperparameters, Their Ranges, and the Best Fit Values

value	range[a]	best fit	comments
training data set size	100−10000[a]	10000	maximum number of valid images
learning rate	10^{-4} to 10^{-2} [a]	10^{-4}	lowest before diminishing returns
batch size	5−20	20	highest within memory limit
number of epochs	5−20	20	highest before diminishing returns
number of nodes in the hidden layer	5−100	70	optimal value while still being computationally expedient

[a]Denotes those that are chosen randomly logarithmically rather than linearly.

For each set of parameters, the MECSim software package[18] is used to generate potential−current−time data which are displayed as current versus potential (i−E plots) and converted into input for the DNN. It is worth noting that MECSim has a key advantage over commercially available software (e.g., DigiSim and DigiElch)[19] in that it can be automatically scripted to run thousands of sets of parameters rather than requiring setting with each parameter set. Figure 1 illustrates the conversion process for the E, EE, and EC mechanisms with the first row showing the current response as a function of the applied potential ramp varying from 0.5 V to −0.5 V and back to 0.5 V. The cyclic voltammograms are then converted into 100 × 100 pixel greyscale images which are normalized to have minimum value of 0 and maximum of 1 and are used as input for the DNN, as shown in the final row.

This 2D image representation for the DNN has two key advantages over a 1D vector representation of the current as a function of time, often used for parameter optimization. First, it allows for multiple convolution layers to run over the image, picking out trainable subshapes, such as edges and curves, that

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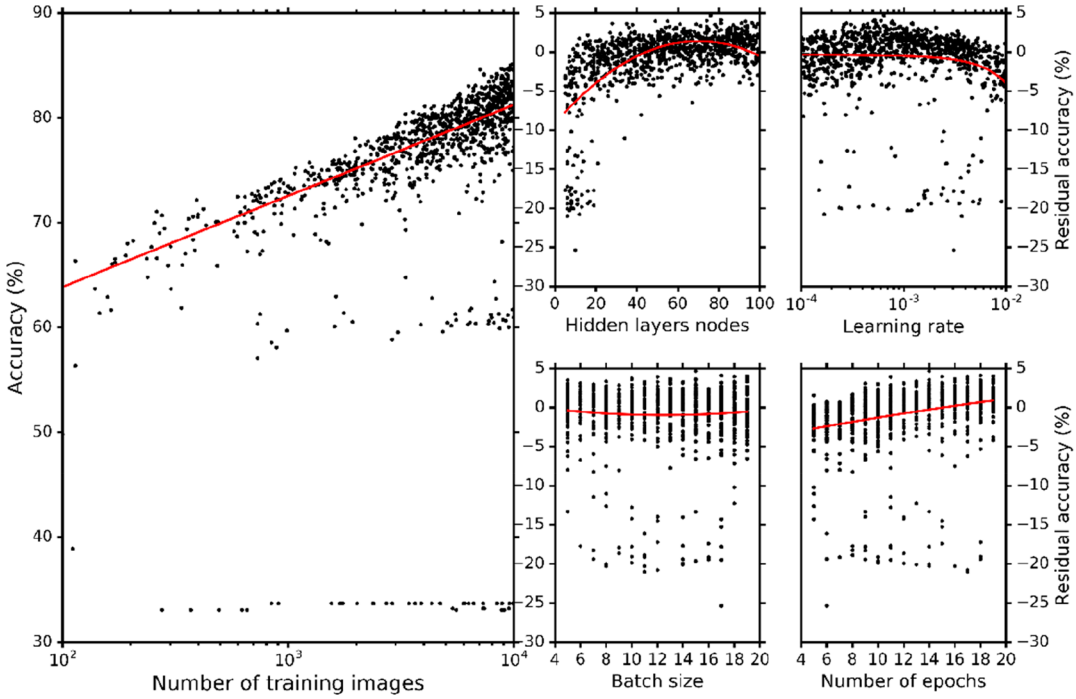


Figure 3. Effect of hyperparmeters on the accuracy determined by randomly sampling each parameter 1000 times. Left panel shows the accuracy against the number of training images used to train the DNN, which is by far the dominant effect. Right panels show the residual accuracy (after subtracting the best fit to the training images) for the other hyper parameters. Fits in red are log−linear for the number of training images and an order 2 polynomial for the residuals for other hyperparameters.

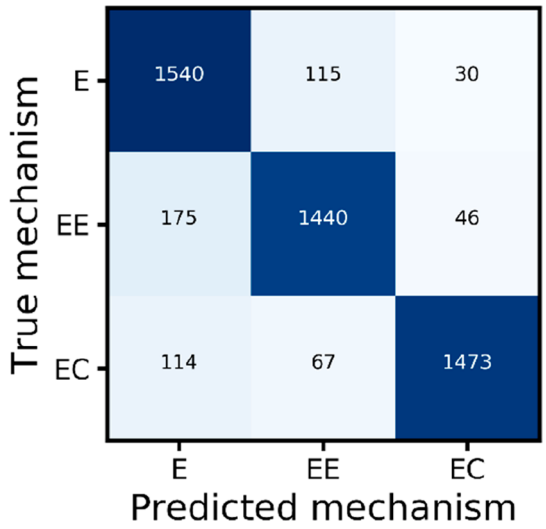


Figure 4. Confusion matrix for the DNN mechanism classifier using the optimal hyperparameter values on the 5000 test images, yielding an overall accuracy of 89%.

together indicate the final classification. Second, this process mirrors how a human goes about classifying an image in that they look for large scale structures, e.g., the gradients of the peaks and the number of peaks, etc., in a holistic way.

The greyscale normalization means that the image is darker in regions with a smaller gradient of change in current magnitude (i.e., at the initial and final potentials) and lighter in the regions where the gradient is steepest. The motivation for this representation is to have available a method of capturing fine detail, such as closely spaced peaks with a low-resolution image required for efficiently training a DNN. Note that when using the original data without this shading, the method tended to overemphasize the steep rises around the peaks, leading to more misclassifications as an EE mechanism. This is unsurprising, because DNN is based on image recognition. Computers "see" an image differently from human beings. However, we expect this difference to become smaller once the extent of training data increases.

For each mechanism, 5000 simulations were run using MECSim, and the resulting cyclic voltammograms were converted into DNN input images. These images were divided into a training and testing set; the training data were used to construct the DNN and the testing data (randomly chosen 5000 images) used for evaluation.

The freely available Python libraries TensorFlow[20] and Keras[21] were used to construct the DNN which is based on a simple handwriting image recognition setup. The DNN architecture is shown in Figure 2 which has a variable learning rate, number of hidden nodes, number of images used for training, batch size, and the number of images per epoch. The Python code for this DNN architecture is given in the Supporting Information (Figure S1). These hyperparameters were tuned to optimize the accuracy of the predictions to the known mechanism using the testing data set. The specific hyperparameter values for the DNN are given in Table 2 where each model took about 4 min to train and less than 5 ms to classify an image (excluding the time taken to load the image and libraries). The number of images used to train the DNN is the dominant hyperparameter in affecting the accuracy of the classifier as evident from Figure 3. In general, the more training undertaken with the DNN method, the better the accuracy. This is not surprising because the analogous comment is also applicable to humans where the highest accuracy is expected to be achieved by the electrochemist with the most experience in the discipline.

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The performance of the highest accuracy classifier is shown as a confusion matrix in Figure 4, which shows the true mechanism used to generate the image on the *y*-axis and the mechanism predicted by the classifier on the *x*-axis. A perfect classifier would have the diagonal in the confusion matrix summing to 5000, i.e., the classifier correctly predicted the mechanism for all test cases. The DNN correctly classifies all three mechanisms in most cases as is evident in most cases falling on the diagonal of Figure 4. Misclassifications between all three mechanisms are discussed along with the causes in the next section.

DISCUSSION

Overall the DNN performs very well, classifying the mechanism with 89% accuracy. However, it is significantly more successful than this implies when the reasons for misclassifications are examined. In short, the misclassifications the DNN makes are exactly what a human does and what is expected from the voltammetry for each mechanism under certain parameters.

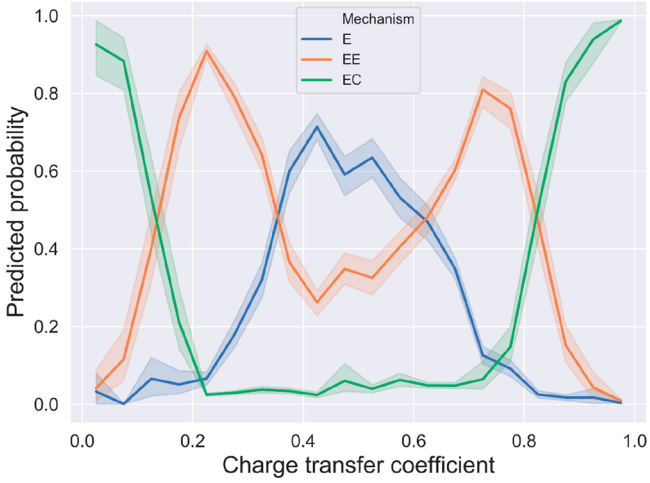


Figure 5. Predicted probability for an E mechanism as the charge transfer coefficient (α) is varied. The simulated data for the E mechanism uses the default parameters given in Table 1 except for α. Confusion with an EE mechanism occurs when α is outside of the typical range of $0.4 < \alpha < 0.6$. When α is 0 or 1, the forward/backward charge transfer reaction is completely suppressed; thus, the cyclic voltammetric patterns converge closely to those expected for an EC mechanism.

Figures 5−8 show three cases where the DNN predicts the correct mechanism under typical conditions but misclassifies the mechanism when the parameters become atypical. Each of these figures shows the mean probability and associated 95% confidence interval around the mean of classifying a particular mechanism against a single variable. For all figures, a total of 1000 simulated data sets were created and split into 20 bins along the *x*-axis. Note that the raw classification outputs do not always add to unity, for example when no classification is found that well fits the input. When the raw classifications are converted to probabilities, they are normalized such that they sum to unity.

For single charge transfer reactions, the charge transfer coefficient (α) gives the degree of asymmetry between the forward and backward reaction where $\alpha = 0.5$ is symmetrical. As seen in Figure 5, the DNN correctly classifies the E mechanism correctly when α is close to 0.5 but misclassifies it as an EE mechanism for moderately asymmetrical reactions and as the EC mechanism for extremely asymmetrical charge transfer reactions. In practice, α values are usually close to 0.5.[17]

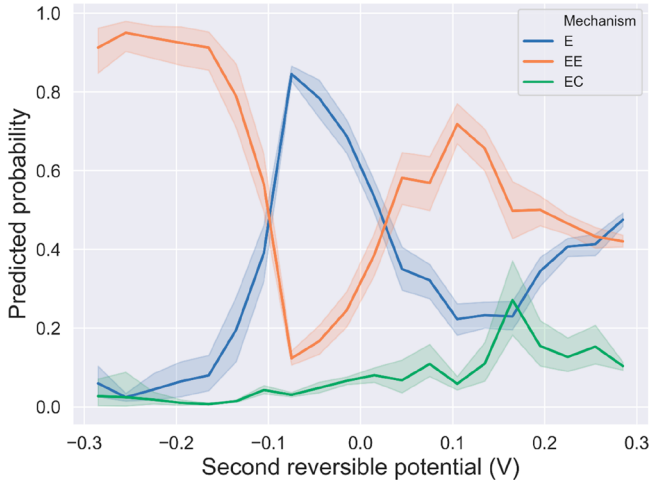


Figure 6. Predicted probability for an EE mechanism as the reversible potential for the second reaction (E^0_2) is varied. The simulated data for the EE mechanism uses the default parameters given in Table 1 except for $k^0_2 = 10^{-3}$ cm s^{-1} and E^0_2. When the peaks are clearly distinguishable at $E^0_2 < E^0_1 = 0.0$, then the EE mechanism is correctly classified. Some ambiguity exists in other regions when the peaks overlap, and EE reduces to a narrower version of an E mechanism.

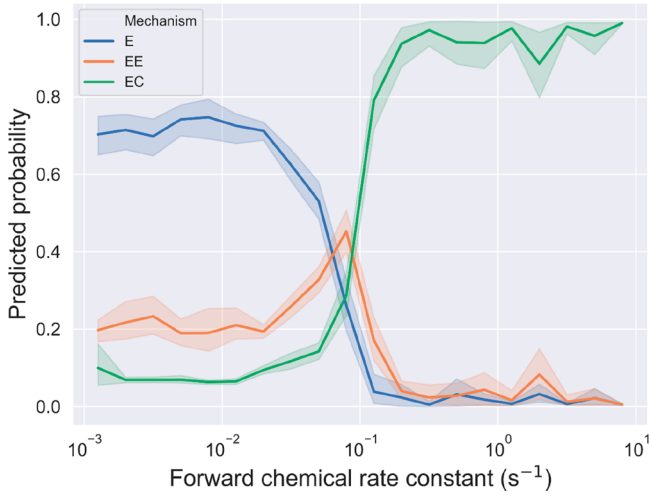


Figure 7. Predicted probability for an EC mechanism as the rate constant k_f is varied. The simulated data for the EC mechanism uses the default parameters given in Table 1 except for k_f. As expected, the classifier sees this as a single electron transfer mechanism when the rate constant for the chemical reaction goes to zero. When the rate constant is significant, then the classifier performs extremely well.

As seen in Figure 6, for an EE reaction, the DNN correctly classifies the input when the peaks are well separated, i.e., the reversible potential for the second reaction $E^0_2 < -0.1$ V. However, it favors an E mechanism when the peaks overlap, which is unsurprising because a human expert could make the same classification under this circumstance. In fact, the theory predicts that reversible E and EE processes are indistinguishable when $E^0_2 - E^0_1 = 35.6$ mV.[22] When E^0_2 is significantly larger than E^0_1, the peaks become narrower than a typical E

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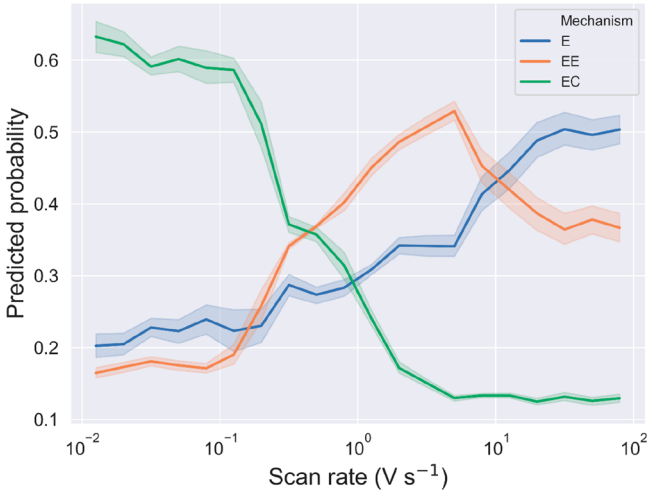


Figure 8. Predicted probability for an EC mechanism as the scan rate is varied. The simulated data for the EC mechanism uses the default parameters given in Table 1 except for k_f = 1 s^{-1} and ν. As expected, the classifier currently identifies the chemical reaction when the scan rate is sufficiently slow for it to affect the shape of the voltammogram. When the scan rate outpaces the chemical reaction kinetics, then it, correctly, confuses the voltammogram with that for an E mechanism.

reaction, which removes the ambiguity once again and the classifier correctly returns EE but with less confidence.

In the final examples, the origin of misclassification is even clearer. Namely for an EC reaction, the system physically reduces to the E mechanism when the forward rate constant of the chemical reaction becomes too slow (Figure 7). As expected, under this condition, an EC reaction transforms into a simpler E reaction on the measurement time scale.[23]

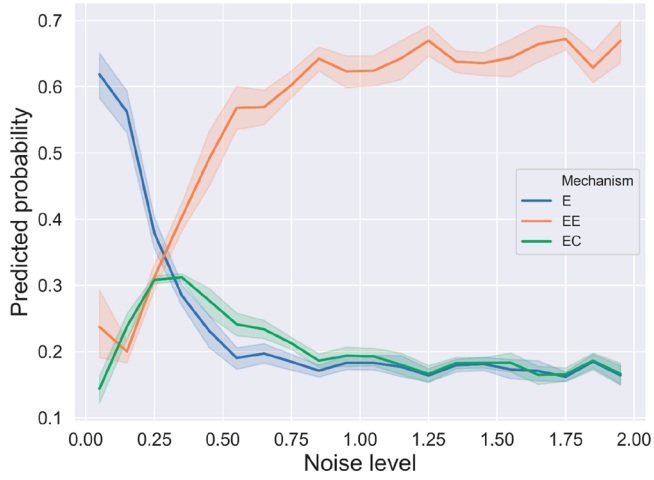


Figure 10. Effect on the classification probabilities of increasing the noise added to the simulated data for an E mechanism shown in Figure 9.

Similarly, this transformation occurs when the scan rate is increased such that it outpaces the chemical reaction (Figure 8).

So far, the concept of using a DNN to classify electrochemical mechanisms has been tested with simulated data. This is hugely advantageous, as it allows us to conveniently sample a large parameter space which is not available with experimental data. However, experimental data contain noise. To test their impact, we added noise to an example of each mechanism using a normal distribution with mean zero and standard deviation given as the scatter factor times the mean of the absolute value of the current. Thus, a scatter of 1 means that the noise level is the same amplitude as is indicative of the

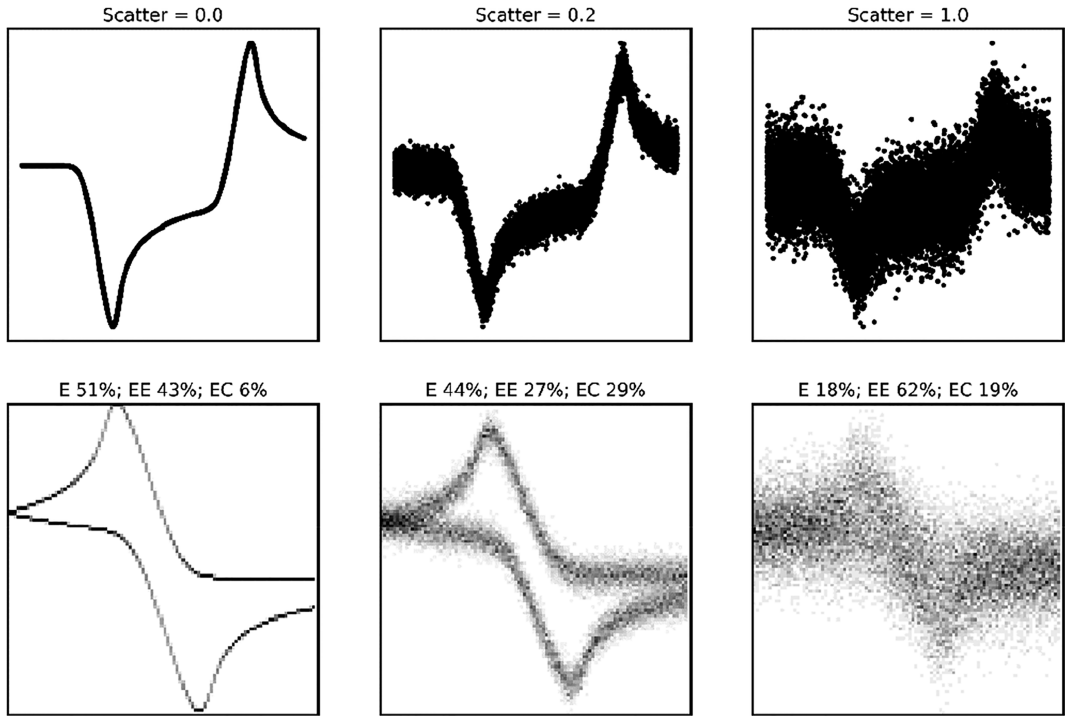


Figure 9. Impact of adding noise to the current as a function of time (top panels) for an E mechanism example on the DNN input images (bottom panels). The simulated data for the E mechanism before noise is added uses the default parameters given in Table 1 except for R_u = 50 Ω and k^0 = 1 cm s^{-1} which were chosen to further confuse the classifier.

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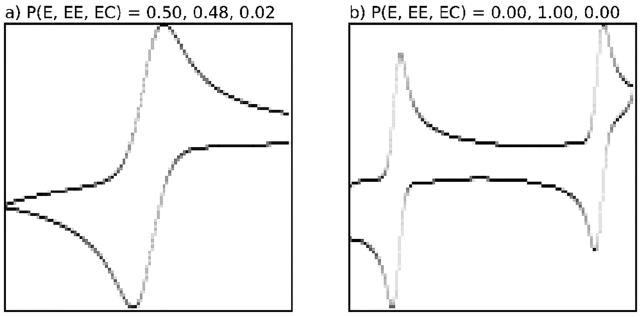


Figure 11. Classification probabilities determined by DNN using experimental cyclic voltammetric data sets derived from (a) E ($Fc^{0/+}$) and (b) EE ($[SVW_{11}O_{40}]^{3-/4-}$ and $[SVW_{11}O_{40}]^{4-/5-}$) mechanisms.

current amplitude. This can be seen in Figure 9 where the top panels show the effect on the current as a function of time for scatter factors of 0, 0.2, and 1 while the bottom panels show the input for the DNN for an E mechanism. A similar consideration applies to the EE and EC mechanisms shown in the Supporting Information.

The effect of increasing the noise from zero to twice the indicative amplitude of the current is shown in Figure 10. As the noise increases, the image that the DNN takes as input looks increasingly like previous examples of EE mechanisms, i.e., multiple peaks taking up much of the image. This results in very noisy inputs being classified as EE mechanism irrespective of the true mechanism. Inclusion of a "too noisy" classification could be undertaken in the future to allow for the extreme noisy data to be excluded from analysis. Human analysis of noisy data would also accept scatter of 0.2 but not 1.0 and probably lead to a recommendation to repeat the experiments. This work reports the first step of automated analysis strategy we are developing that will take in experimental data, recognize the broad mechanism (e.g., 50% chance of an E and 30% of an EC, etc.), and then optimize the parameters of each of these to determine better fits to the data.

As a final test for the applicability of the DNN method, the classifier was applied to two sets of experimental cyclic voltammetric data. Figure 11 provides the classification probabilities above the diagram displaying what the DNN takes as input from the experimental data used for mechanism E (the $Fc^{+/0}$ (Fc = ferrocene) process obtained with a 1 mm diameter glassy carbon electrode in a $BMIMPF_6$ solution containing 5 mM Fc at a scan rate of 0.1 V s^{-1}) and EE (Figure 2a in ref 24 associated with the $[SVW_{11}O_{40}]^{3-/4-}$ and $[SVW_{11}O_{40}]^{4-/5-}$ processes at a GC electrode in dimethylformamide). In neither case is the EC mechanism seen as a probability. For the EE mechanism (right panel) case, the DNN trained on simulated data gives the correct classification as the image is perfectly clear. The same conclusion would be reached by an electrochemist examining this cyclic voltammogram. For the E mechanism case shown in the left panel, the outcome of the DNN exercise is uncertain as to whether the mechanism is E or EE, as again would apply an experienced electrochemist examining the data in the absence of additional information. This is due to the limitation associated with dc voltammetry, in which an EE mechanism with overlapping peaks is under some circumstances almost identical in shape to an E mechanism (see Figure 6). Thus, unambiguous distinction for both the classifier and human requires more training data or information. This ambiguity will be cleared up in a future version of this work which will be extended to include inputs from, for example, the fundamental and higher order harmonics in Fourier-transformed ac voltammetry.[18,25]

CONCLUSIONS

We have successfully trained a deep neural network on dc cyclic voltammograms for three commonly encountered mechanisms and shown that it not only successfully classifies the mechanisms but that misclassifications are due to well-known problematic combinations of the physical mechanism parameters. This method replaces the current subjective practice of an experienced electrochemist recognizing the mechanism based on the cyclic voltammogram with the bonus that the confidence and potential for alternative mechanisms are quantified. While dc cyclic voltammograms have been used in this report, other forms of dynamic voltammetry could be even more powerful when utilized with the DNN approach. For example, ac voltammetry[2,25] provides frequency and harmonic dependencies which convey substantial additional information to that available in the dc method.

A DNN classifier of the kind generated in this study is a critical component of any future automated data analysis beginning with the raw experimental data and ending in a ranked order of mechanisms with associated optimally fit parameters. We see no reason why the successful approach introduced here would not advance the practice of voltammetry and become a complementary tool to mathematic modeling as in other research fields, such as batteries.[14]

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.analchem.9b01891.

Python code for the DNN as well as data showing the impact of noise on the classification of EE and EC processes (PDF)

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Notes

The authors declare no competing financial interest.

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